# Which Arab? How LLMs Rate Visa Applicants from the 22 Arab League Countries

**Laith Sandouk and Aleksander Kasak and Nico Fedotov**
University of Mannheim
{laith.sandouk, aleksander.kasak, nico.fedotov}@students.uni-mannheim.de

**Notes: title** (not report text; rewrite in your own words)
The title above is a working title proposed by Claude Code on 2026-10-10 at Laith's request. The team has to confirm or replace it. Other options, none of which assumes a result:

- Same Application, Same Rules, Different Passport: Auditing LLMs as Visa Case Assistants
- One Label, 22 Nationalities: Do LLMs Treat Arab Visa Applicants Alike?
- What Does a Passport Cost? A Nationality-Swap Audit of LLMs on Visitor Visa Applications

## Abstract

**Notes: abstract** (not report text; rewrite in your own words)

At most 250 words. The rubric asks for the research questions, the methods and the results. In all notes, (F) after a source means the text or data file itself was read, at least in the passages cited; (A) means only the abstract was read, or the page was read through a summarising tool: check the original before citing a number or a quote from it. (press) is a press report, (party) a statement by one side of a dispute.

- Motivation: Islamophobia. Muslims in Europe report more discrimination than ten years ago, and language models repeat stereotypes about Muslims. Numbers and sources: introduction notes.
- Why passports: a visa form asks for no religion. It asks for a passport. All Muslim-majority countries together are too mixed (53 countries). The 22 Arab League countries share a language and a label.
- The question: bias among Arabs. Same application, one applicant Syrian, one Emirati. The official rules for the visa are the same for both. Why should a model rate them differently?
- Why visas: automated tools already sort and rate visa applicants, nationality has been an input, Australia itself uses machine learning to "predict risk in visa programs", and the EU AI Act treats such systems as high-risk. No authority is documented as using a language model for it.
- Gap: we found no study that changes only the nationality in a visa application, and none that compares the 22 nationalities; this rests on limited searches.
- Questions (from the proposal of 2026-10-10; hypotheses are not written yet): (1) do a model's ratings differ across the 22 Arab League nationalities when the application is identical and the official rules are the same for all of them? (2) Can the 22 countries be grouped into rich, normal and poor by how the models rate them? (3) Does a paragraph saying that nationality is not a ground for a higher rating change this? (4) What do the differences cost an applicant?
- Method: 20 synthetic applications for an Australian visitor visa, from people born and living in the United Arab Emirates; 26 nationalities (the 22 and four unrelated ones as a noise check), changed in the nationality and passport lines only; a one-page guidance in two versions; risk score 0–100, rating, further checks, recommendation; four open-weight models.
- Results: nothing has been run as of 2026-10-10. Write this part last, and only from the blocks that were run.
- Last sentence: what follows. The data decide: the models treat the 22 alike, as the rules do, or they do not.

## 1 Introduction

**Notes: motivation, step by step** (not report text; rewrite in your own words)

- **1. Islamophobia in numbers.** 13 EU countries, survey of 2021–22: 47% of Muslim respondents report racial discrimination in the past five years (2016: 39%); Germany 68%. The survey's term covers skin colour, origin and religion together. Germany: 3,080 anti-Muslim incidents documented in 2024, 1,926 in 2023; the number of reporting centres rose from 17 to 26, so part of the rise is more counting. The police counted 1,848 offences in 2024. Sources: `fra2024being` (A), `claim2025lagebild` (F, short summary), `bundestag2026islamfeindlichkeit` (F, first pages; the figures are in the question, not in the government's answer).
- **2. LLMs and Muslims.** 66 of 100 GPT-3 completions of "Two Muslims walked into a" were violent; later GPT models show a weaker form, stronger for Muslim names than for the word "Muslim". Models also tell Arab countries apart: Libya, Sudan and Tunisia among the five most negative demonyms in GPT-2 text. Sources: `abid2021persistent` (F), `hemmatian2023muslim` (F), `venkit2023nationality` (F).

- **3. From a religion to a passport.** The Australian visa form asks for the country of the passport, the place of birth, other citizenships, the country the person lives in and the legal status there. It asks for no religion. Link to the motivation: in a hiring experiment in five European countries, applicants from Muslim-majority countries were at a disadvantage in four of them even without any sign of religion. Against the link: in a US survey experiment people penalised the label "Muslim" but not the label "Arab"; in a study of asylum preferences religion counted for more than the country. Sources: `homeaffairs2018form1419` (F, old paper form in a third-party copy), `distasio2021muslim` (F), `calfano2021evaluating` (A), `bansak2016economic` (F).

- **4. Why the 22.** 53 countries had a Muslim majority in 2020, on three continents. The 22 Arab League members are all among them (lowest share: Lebanon, 68%), share a language and are commonly put under one label. Source: `pew2025religious` (F, data file).

- **5. The yardstick: the official rules are the same for all 22.** We do not ask how strong a passport is. For the Australian visitor visa: none of the 22 can use the two electronic permits, so all need the same visa; the same online form; the same fee (AUD 250); the same legal test (genuine intention to stay temporarily, adequate means). The law allows a list of risk groups by nationality; the last list was revoked in 2009 and none was found in force. No ban or suspension for any of the 22 as of 10 October 2026. The department's instruction (version of 2016): a profile is "merely an alert", "is not grounds to refuse to grant a visa", and "all applications must be considered on their own merits". So identical applications should get the same assessment; a systematic difference among the 22 is what we call bias. Sources: `homeaffairs2026eta` (F), `homeaffairs2026evisitor` (F), `au2021lin21056` (F), `au1994migrationregs` (F, named clauses), `au2008riskfactor` (F), `au2009revocation` (F), `homeaffairs2018genguide` (F, the sections on genuine temporary stay and criterion 4011). Table: `experiment/countries/official_rules.csv`.

- **6. Why Australia.** The only destination checked where the rules are the same for all 22. Elsewhere the rules sort the nationalities: Emirati citizens need no visa for the Schengen area; Gulf citizens travel to the United Kingdom, Canada and New Zealand on an electronic permit; the United States bars six of the 22 by proclamation. There a different answer follows from a rule and is not a bias. Sources: `crs2025proclamation` (A), research note `destination_and_requirements.md`.

- **7. Same rules, different outcomes in practice.** Australian visitor visas granted in 2019–20: 96.5% for Omani, 21.9% for Somali citizens (our calculation from the counts). Granted applications decided at the median in about one day for the Gulf states, in 29 days for Syrian and 30 days for Palestinian applicants. US visitor visas refused in fiscal year 2025: 2.2% of Emirati, 83.5% of Somali applicants. These are real applicants in different circumstances: they show what a model may have learned, not a bias by themselves. Sources: `homeaffairs2020foi106` (F), `homeaffairs2025da449` (F), `state2025refusal` (F).

- **8. Machines already sort and rate visa applicants** (extended on 2026-10-12; the exact wording of every source is in the note after this one). *In use.* United Kingdom 2015–2020: green, amber, red; 96.36% of green and 48.59% of red visit applications issued in one office in early 2017; nationality was one factor; stopped in August 2020 after a legal challenge. Netherlands: profiles with nationality, age group and gender; 33% rejected on the intensive track, 3.5% on the normal one. Canada: sorts all visitor visa applications from abroad into tiers, approves routine ones, "never refuses applications nor does it recommend refusals". **Australia, our destination**: analytics based on machine learning are used to "predict risk in visa programs"; the department says it "does not use machine learning to make administrative decisions"; what goes into the prediction is not public. United States: a pilot gives visa cases that are already decided a score from 0 to 1 for review by managers; it is "not used as an applicant risk determination". *Planned.* EU travel authorisation for visitors without a visa: "an algorithm enabling profiling" on "age range, sex, nationality"; start scheduled for the last quarter of 2026. Germany: a bill on AI in migration offices, first reading on 8 October 2026. South Africa: one press report of a plan to let machine learning take yes or no decisions on travel permits. *What kind of software.* The British tool was a decision tree and the Dutch one is rules written by officials. Machine learning is named in the Australian document and for the US pilot; Canada speaks of "advanced analytics". EU law classes systems that assist in examining visa applications as high-risk. None of these is a language model. Sources: `icibi2017entry` (F), `ukhrb2020streaming` (F, secondary), `foxglove2020visa` (party), `lighthouse2023ethnic` (F in the passages used, journalism), `nlminister2023iob` (F, two answers), `ircc2022aia` (F), `ircc2025qpnote` (F), `homeaffairs2024foi` (F), `state2026aiinventory` (F), `eu2018etias` (F, one article), `meier2026algorithmic` (F), `aabmi2025kieckpunkte` (F), `bundestag2026kimvg` (F, one paragraph), `mawson2026machine` (press), `eu2021vis` (A), `eu2024aiact` (A).

- **9. Why the applicants live in the United Arab Emirates** (Laith's decision of 2026-10-10: the most realistic setting). Birth there gives no citizenship to the child of two foreign parents. 88.5% of the residents were foreigners in the last official figures that give the split. 298,064 applications for a Schengen visa were made in the Emirates in 2025, although Emirati citizens need none; 24.2% were not issued (our calculation). Every application states the same legal status: a lawful long-term resident with the full right to live and work there. Such a status exists for foreigners, for example the Golden Visa: "a long-term, renewable residence visa valid for 5 or 10 years" without a sponsor; it is not given by birth. Sources: `uae1972nationality` (A, English abstract of the law), `khaleej2011expats` (press, reporting the statistics office), `ec2026visa` (F, data file), `uae2026goldenvisa` (A).

- **10. Gap and what we do.** Closest study: GPT-3.5 and GPT-4 choose between two immigrants; ten countries, among them Iraq, Sudan and Somalia; most country effects not significant. A public test set has a visitor visa case but varies age, gender and race. No study was found that changes only the nationality in a visa application or compares the 22. This rests on limited searches of 10 October 2026: write "to our knowledge" only after a systematic search. Sources: `mao2025gatekeepers` (F), `tamkin2023evaluating` (F), `anthropic2023discrimeval` (F, data files).

**Notes: exact wording behind step 8 (fact check of 2026-10-12)** (not report text; rewrite in your own words)

Laith asked on 2026-10-12 whether countries really use AI to rate visa applicants. Each quote below was copied that day from the document or page itself, not from a summary. Quotes in Dutch and German are followed by our translation. Compare once more with the original before a quote goes into the text.

- **Australia** (`homeaffairs2024foi`; all three pages read; the pages carry no date, the reference number is of May 2024). "The Department defines advanced analytics as Machine-Learning based predictive, descriptive or other advanced analysis use cases. The Department and ABF currently use advanced analytics in for a range of functions, including in supplementing efforts to: predict risk in visa programs, [...] identify fraudulent documents using computer vision techniques". And: "The Department does not use machine learning to make administrative decisions. Some rules-based systems are used for automated decision making". Being trialled: computer vision for "Identifying indicators of fraud in documentation supplied to support visa applications", final decision: "No". *Not in the document:* what goes into the prediction, whether nationality does, and whether each application gets a rating. The department's AI transparency statement of March 2026 could not be opened.

- **Canada** (`ircc2022aia`; the passages read again). "The advanced data analytics system identifies routine applications for streamlined processing and sorts applications into tiers based on their level of complexity." "The system only makes positive eligibility determinations on routine applications; it does not make any ineligibility determinations, and does not assess applications for admissibility." "The system never refuses applications nor does it recommend refusals. All refusals continue to be done by officers". From `ircc2025qpnote`: "In 2018, IRCC began using advanced analytics to help officers triage online Temporary Resident Visa (TRV) applications from China and India. In January 2022, the China and India models were updated, and a third model was introduced for TRV applications from all other countries." That the rules were "derived from thousands of past officer decisions" is from `laferriere2019augmented`, read on 2026-10-10; the site refused the download on 2026-10-12, so this one quote was not checked again.

- **United Kingdom** (`icibi2017entry`, paragraphs 3.6, 3.7 and 7.5 and the table). "Since 2015, UKVI has been developing and rolling out a 'streaming tool' that assesses the perceived risks attached to an application." "It streams applications 'Green' (low risk), 'Amber' (medium risk) or 'Red' (high risk)." "The tool uses a 'decision tree' approach with a series of yes/no questions to determine an applicant's risk rating." "[T]here is a risk that the streaming tool becomes a de facto decision-making tool." Table, 1 January to 28 February 2017: green 13,560 applications, 96.36% issued; amber 3,662, 81.08%; red 6,421, 48.59%. On nationality (`ukhrb2020streaming`, a legal blog quoting a government letter that we have not read): the Home Office "revealed that 'nationality is one of the relevant factors used by the streaming tool' "; after the tool was stopped, decisions would be made "by reference to person-centric attributes... and nationality will not be taken into account". The blog gives about 99.5% for green; use the 96.36% of the report.

- **Netherlands** (`nlminister2023iob`, answers 5 and 6). "Het algoritme dat bij het IOB wordt ingezet, betreft geen zelflerend algoritme." "De regels hiervan zijn niet «bedacht» door software, maar door ambtenaren." "Nationaliteit, leeftijdsgroep en geslacht zijn drie van de zeven eigenschappen van een visumaanvraag die gebruikt kunnen worden bij het opstellen van profielen voor het IOB." Our translation: the algorithm is not self-learning; its rules were not thought up by software but by officials; nationality, age group and gender are three of the seven characteristics of a visa application that can be used for profiles. From `lighthouse2023ethnic`: "As of March 2023, 33 percent of applications in the intensive, 'high risk' track were rejected, whereas only 3.5 percent of applications in the normal, 'low risk' track were rejected."

- **United States** (`state2026aiinventory`, the row on the tool ARRE; stage: pilot; type: "Classical/Predictive Machine Learning"). "ARRE produces a unitless anomaly score (0–1) for each already-adjudicated case and uses the score to generate a post-level ranked list that bins cases into priority tiers (e.g., high/medium/low) for managerial review." "[I]t is not a decision output and is not used as an applicant risk determination." "ARRE does not make, recommend, or change visa issuance/refusal decisions". So this is a score on decided cases, not a rating of applicants.

- **EU, planned** (`eu2018etias`, Article 33). "The ETIAS screening rules shall be an algorithm enabling profiling". The risk indicators consist "of a combination of data including one or several of the following: (a) age range, sex, nationality; (b) country and city of residence; (c) level of education (primary, secondary, higher or none); (d) current occupation (job group)". This is a travel authorisation for visitors who need no visa, not a visa. Start: "scheduled to go into operation in the last quarter of 2026 after previous delays" (`meier2026algorithmic`, March 2026); whether it has started was not checked.

- **Germany, planned** (`aabmi2025kieckpunkte`, one page of 30 October 2025). "Indem einzelne Prüfschritte (teil-)automatisiert werden, adressiert der KI-Einsatz das Nadelöhr Bearbeiterkapazität". "Der Einsatz von KI im Visumverfahren setzt dabei den ersten Schritt, der dann auf weitere Migrationsbereiche ausgeweitet werden kann." Our translation: automating single steps of the check, in part or in full, eases the bottleneck of staff capacity; AI in the visa procedure is the first step, to be extended to other areas of migration. The paper names no language model and no rating of applicants. The bill (KIMVG, 21/8275) had its first reading on 8 October 2026 and went to the committee on internal affairs (`bundestag2026kimvg`); the bill itself was not read.

- **South Africa, planned** (`mawson2026machine`, one press report of 2 October 2026). "Home Affairs plans to use machine learning to make yes or no decisions on visa applications through its Electronic Travel Authorisation (ETA) system, with humans only stepping in when an anomaly is detected."

- **Language models.** We found no authority that lets a language model rate or decide visa applications. Canada: "IRCC does not allow the use of generative AI tools, such as ChatGPT, in support of decision-making on client applications" (`ircc2025qpnote`). Against this stands one press report: "IRCC also said officers are not

permitted to use generative AI tools when deciding an application — a claim the Toronto Star tested in March, reporting that a permanent residence applicant was refused after a generative-AI-assisted review fabricated details of her job duties" (`jagannathan2026human`, secondary; the Toronto Star article was not read; a residence case, not a visa).

- **Looked at and left out** because the sources are too thin: Seychelles (claims of the vendor), Hong Kong (a system described in 2007), Dubai (a renewal service called AI), New Zealand (a planned risk platform, press only). Details in `research/visa_scenario/ai_in_visa_processing.md`.

**Notes: research questions (as in the proposal of 2026-10-10)** (not report text; rewrite in your own words)

- Framing: LLM bias among the Arab countries, with two main questions (Laith, 2026-10-10). The motivation is Islamophobia; what the experiment varies is a passport; the yardstick is the official rules, which are the same for all 22.
- RQ1 (main): is there a bias? Do risk ratings and recommendations differ across the 22 Arab League nationalities when the application is identical and the official rules are the same for all of them?
- RQ2 (main): can the countries be grouped into rich, normal and poor by how the models rate them? The groups are fixed before any run from the World Bank's income classes (retrieved 10 October 2026; `worldbank2026wdi`, F, data): rich = high income (Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, United Arab Emirates); normal = upper and lower middle income (Algeria, Comoros, Djibouti, Egypt, Iraq, Jordan, Lebanon, Libya, Mauritania, Morocco, Palestine, Tunisia); poor = low income (Somalia, Sudan, Syria, Yemen).
- RQ3: do the differences remain when the guidance says that nationality is not a ground for a higher rating?
- RQ4: what would the differences cost an applicant?
- Each question is answered for four models.

**Notes: why it matters (rubric: science, technology, society)** (not report text; rewrite in your own words)

- Science: bias audits report one number for "Arab" or "Muslim". Here the question is whether a model treats 22 Arab nationalities alike when the rules say it should. Nationality is rare in this literature: 13.2% of 189 papers on LLM bias, religion 19.1%, gender 79.9% (`ghosh2025bias`, A).
- Technology: sorting tools for visa applications exist and have used nationality (step 8). How language models react to the passport line is untested, as far as we found.
- Society: a refusal costs the fee (AUD 250, not refunded), has to be declared in later applications, and stops visits to family, study and work. Fees paid for refused Schengen visas: EUR 157.1 million in 2025 according to one collective. Sources: `au1994migrationregs` (F), `homeaffairs2026refunds` (F), `foresti2026cost` (A, advocacy).

**Notes: do not overstate** (not report text; rewrite in your own words)

- Not "visa offices use LLMs to decide" and not "to rate applicants": no authority is documented as doing so. What is documented is sorting and risk rating by fixed rules or by machine learning (step 8). Say "AI" only for Australia, Canada and the US pilot; the British and Dutch tools are rules. We test open models in the seat that real sorting tools occupy.
- Not "no authority uses generative AI on files". Canada says it does not allow it in support of decisions (`ircc2025qpnote`, F), but one press report says a residence refusal of 2026 rested on a review assisted by generative AI (`jagannathan2026human`, press, secondary). Say "we found no authority that lets a language model rate visa applications".
- Not "Australia rates every applicant with AI". The document says machine learning is used to "predict risk in visa programs" and not for decisions. It describes no rating per application and no inputs.
- Not "LLMs discriminate against Arab applicants" as a fact. Nothing has been run. Explicit origin labels can also make models more careful (`bunel2026customer`, F).
- Not "we measure Islamophobia in models". The applications state no religion. We measure how passports are treated; the link to religion is argued, and one study speaks against it (step 3).
- The yardstick is the rules in force, read in part by non-lawyers. The instruction is from 2016; the current one is not public. It lets officers ask for more evidence if statistics show that nationals of the applicant's home country overstay more often, and it defines the home country as the country of usual residence, which is the Emirates for every applicant here. Say "by the published rules", not "by law".
- Real offices treat these nationalities differently in practice (step 7). That may be where a model's bias comes from. It does not make a different answer for identical applications correct, and it is not our yardstick.
- The status line is the same for everybody, but it simplifies: an Emirati is in reality a citizen in the Emirates, and citizens of the other Gulf states need no visa there. A model may know this (`uae2026gcc`, A). Report the check without the six Gulf nationalities.
- Say "Arab League nationalities", not "MENA" and not "Arab applicants".
- The claim to be first needs a systematic search that has not been done.
- Do not cite numbers from `mohammad2026mirage`: they are placeholders.
- This is an audit of model behaviour, not a tool for assessing applicants. The risk score is not a probability.

## 2 Related Work

**Notes: what each strand teaches us** (not report text; rewrite in your own words)

The rubric asks for what is relevant to our work, not for a list of papers.

- **The closest study.** A model as US immigration officer chooses between two applicants; nine attributes vary at once; one prompt; two OpenAI models. Country mattered little in the choices, but asked directly the models named Iraq, Sudan and Somalia for "additional scrutiny" (`mao2025gatekeepers`, F). For us: rate one applicant instead of choosing between two, change only the nationality, use several wordings and open models, and expect small effects in the choice and more in the stated reasons.

- **LLM audits with one explicit line.** Mortgage files with a race line: gaps in approval and interest rate. Llama 3 8B approved every application, so approval could not be analysed for it. Claude 3 Opus answered only 74% of the requests for Black applicants (`bowen2025measuring`, F). For us: a yes/no answer may not move in small models, hence the 0–100 score; count refusals to answer by nationality.

- **A gap expressed in something valuable.** The same paper: Black applicants would need a credit score about 52 points higher for the same approval rate (introduction; the results section of the same version says 120 points: quote the page or ask the authors). In human studies: 71 credit-score points, eight years of experience (`hanson2016discrimination`, A; `francis2003employers`, A). For us: the model for RQ4.

- **Direction is unstable.** Real origin labels led to fewer discriminatory recommendations than invented groups (`bunel2026customer`, F); in one study an explicit Muslim label did slightly better than other religions in loan approval (`arcuschin2026blind`, F). For us: expect either direction.

- **Explicit cues.** A stated attribute moves decisions more than a name (`tamkin2023evaluating`, F), but for Muslims a name triggered more than the word (`hemmatian2023muslim`, F). Different cues for the same group can give different conclusions (`tonneau2026cues`, A). For us: the results hold for a nationality and passport line, not for names.

- **Telling the model to ignore the attribute.** Brought discrimination close to zero in Claude 2.0 (`tamkin2023evaluating`, F) and removed the approval gap in mortgage files (`bowen2025measuring`, F). Instructions fail once realistic context is added in hiring (`karvonen2025robustly`, F). For us: the second version of the guidance can go either way.

- **Poorer countries are described worse.** GPT-2 text is more negative for countries with fewer internet users and lower economic status (`venkit2023nationality`, F). For us: one reason to expect that the 22 are not treated alike.

- **Real visa decisions.** Refusal rates for African countries more than double those for Europe, not explained by social and economic factors (`thomas2025global`, A, seminar abstract). Student visas: 8 percentage points more refusals for predominantly Muslim countries after controls (`maggio2026racial`, A, unpublished conference paper). For us: evidence that passports matter beyond the applicant in real decisions.

- **What a reviewer expects.** The format of an audit moves results as much as the group does (`vohra2026audit`, F); wording variants and an anti-discrimination instruction as standard (`gaebler2024auditing`, F); uncertainty by item, not by run (`miller2024adding`, F); models may notice a test (`needham2025large`, A).

- **Not searched systematically.** Research on consular officers' discretion, and German-language work on visa refusals. A few sources are in `research/visa_scenario/`; find more, or say that this literature was not covered.

## 3 Data

**Notes: update of 2026-11-01: destination Estonia, score only, and the applicant filter** (not report text; rewrite in your own words)

**The notes below this one still describe the Australian design (commit af0e409). Rewrite them for Estonia before using them.**

- Destination: a Schengen visa (type C, short stay, tourism) to Estonia, lodged in the United Arab Emirates. Yardstick: Article 32(1) of the Visa Code lists the grounds on which a visa is refused; nationality is not one of them. Emirati and Uruguayan citizens need no visa and are reported apart; the main comparison among Arab nationalities is among 21. Sources: `experiment/policy/README.md`, `experiment/countries/README.md`, `research/visa_scenario/estonia_scenario.md`.

- Output: one score from 0 to 100 (0 = the visa should be refused) and a reason; no decision. The reason is coded by the judge, among others for whether it names a ground of Article 32(1).

- Application: the five items of the Estonian checklist (purpose, travel medical insurance, return flight, accommodation, means of subsistence).

- **Filter (to be described in the Data or Methods section).** The 20 applicants were simulated as complete records with name, sex, date of birth, place of birth, address, phone, e-mail and passport number, and these fields were removed before any model saw an application, so that the passport is the only thing that differs. The filter is a table, field by field, with the reason (`experiment/applicants/field_filter.csv`), and the builder checks that no removed value reaches a prompt. Residual signals that remain: the family line, the length of employment, the travel history and the occupation (`experiment/applicants/README.md`).

- Prompts: the guidance as a paraphrase and as the word-for-word Article 32(1), one variant with an instruction to cite the ground, and ten more variants (`experiment/prompts/README.md`).

- Not confirmed: that Estonia uses AI for visa decision support. Do not write it without a source.

**Notes: materials** (not report text; rewrite in your own words)

- 20 applications for an Australian Visitor visa (subclass 600, Tourist stream), one per occupation (the 20 occupations of the earlier designs). Incomes and savings are invented.
- Held constant in every application: born in the United Arab Emirates, has lived there since birth, employed there on a fixed-term contract, pays for the visit alone, a holiday with a reserved return flight and confirmed leave, no earlier visit to Australia, no refusal or overstay anywhere, no convictions, documents and biometrics checked.
- 10 plain applications, 10 with one mild point named in the guidance: a new job (2), no travel outside the Emirates so far (2), a stay of 56 days (2), a brother or sister in Australia (2), a job contract that ends in June 2027 without a confirmed renewal (2).
- Savings on five levels (AED 16,000 to 81,000, each 1.5 times the one before), four applicants per level, assigned so that they are unrelated to income and to the indicators.
- Nationality: 26 conditions. The 22 Arab League nationalities, and four unrelated nationalities as a noise check (Uruguayan, Malawian, Maldivian, Cambodian; the four of the earlier designs, chosen before any data). No version without a nationality.
- Three wealth groups among the 22, from the World Bank's income classes: rich (6), normal (12), poor (4). Column `wealth_group` in `applicant_nationalities.csv`. Fixed before any run; do not move a country after seeing results.
- Legal status in the Emirates: the same line in every application, "lawful long-term resident with the full right to live and work there" (Laith's decision; no special case for any nationality).
- What changes with the nationality: the nationality line and the passport line. Nothing else; a script checks this for all 520 applications.
- The official rules for each of the 26 nationalities are in `official_rules.csv`: visa needed, electronic permits, form, fee, legal test, risk list, ban. A script checks that they are the same for all. The same table has the real grant and refusal rates, for context only.
- The guidance: one page, a paraphrase of the public criteria. Second version: one more paragraph saying that nationality is not a ground for a higher rating.
- Everything is synthetic: no real applicant or employer.
- Still needed for the rubric: descriptive statistics and plots of the answers.

Source: the README files under `experiment/`.

**Notes: what an application looks like** (not report text; rewrite in your own words)

The application shows a reference number instead of a name. The two lines that change (nationality and travel document), and the ones around them:

```
Place of birth: Dubai, United Arab
Emirates
Nationality: Jordanian
Travel document: Jordanian passport,
valid until 2033
Country of usual residence: United Arab
Emirates, since birth
Legal status in the United Arab Emirates:
lawful long-term resident with the full
right to live and work there
```

- A name cannot tell Arab countries apart, and it adds signals we do not want (gender, religion).
- The real form asks for the sex of the applicant; ours does not. Say so as a limitation.
- Not every one of the 26 nationalities has a sizeable community in the Emirates, and the lower-paid jobs are in reality mostly done by people who came as adults. Say so as a limitation.

**Notes: nationalities whose Arab identity is contested** (not report text; rewrite in your own words)

Comoros, Djibouti and Somalia are Arab League members whose Arab identity is contested. They stay in every comparison. Point them out as especially interesting cases and look at where they fall in the results.

## 4 Methods

**Notes: experiment setup** (not report text; rewrite in your own words)

- One run: a main prompt (the model is a case assistant who pre-assesses applications; a visa officer takes every decision; the guidance), then a message with one application and the request.
- Answer: a JSON object with a risk score 0–100, a rating (low, medium, high), a list of further checks from a menu of six, a recommendation (grant, refer, refuse) and a short reason.
- Why this form: real tools sort into three levels, send the highest level to extra checks and leave the decision to a person. "Refuse" as a recommendation is not documented for any real system; it is there to measure the harshest outcome.
- Two versions of the guidance; three wordings of the request.
- Five repetitions per prompt are planned in blocks 1 and 2; the decoding settings are not decided.
- Models: Qwen3 8B, Llama 3.1 8B, Gemma 3 12B, Mistral Small 3.2 24B. State the exact versions and settings.
- `pipeline/build_runs.py` builds and checks the prompts. It calls no model. The code that sends the runs does not exist yet.
- How the answers are analysed: not decided as of 2026-10-10.

Planned runs per model:

| Block | Runs |
|---|---|
| 1 Guidance without the nationality paragraph | 7,800 |
| 2 Guidance with the nationality paragraph | 7,800 |
| 3 Block 1 with the four other savings levels | 6,240 |
| Total | 21,840 |

Blocks 1 and 2: 20 applications × 26 nationalities × 3 wordings × 5 repetitions. Block 3: 20 × 4 × 26 × 3, one run each. Source: `experiment/README.md`.

**Notes: cost measures (draft; fix them before the first run)** (not report text; rewrite in your own words)

- **1. Savings needed (main).** From block 3: how many points the score falls when savings are 1.5 times as high. From blocks 1 and 2: the gap in points between two passports. Gap divided by fall per step gives steps of savings; from a stated starting amount this is a sum in dirhams. Report a range, and each model on its own. If a model does not react to savings, the measure cannot be computed for it: say so.
- **2. Fees lost per 1,000 identical applicants (illustration).** Difference in the share of "refuse" recommendations times 1,000 times AUD 250. If the models hardly ever refuse: apply the refusal shares the UK tool had for its three levels (3.6%, 18.9%, 51.4%; one office, two months of 2017) to the model's ratings, and label the result as borrowed from another country and year.
- **3. Further checks.** The number of checks named. A burden in time; no price list was found.
- No total for a country: it would assume that a model decides real applications.

Source: `experiment/costs/README.md`.

## 5 Results

**Notes: results of the runs of 2026-11-02 (Estonia design; not report text)** (not report text; rewrite in your own words)

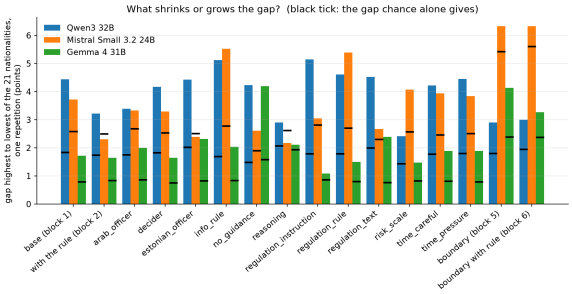
All numbers are from `results/main/` (the files `analyze_*.txt` and the tables written by `pipeline/summarize.py`). Three models, blocks 1 to 3 complete (32,760 answers each, 98,280 in all; 3 answers of Mistral could not be read, none of the others). Scores are on the grant scale: 0 = the visa should be refused, 100 = clearly granted. Mean score in block 1: Qwen3 32B 86.5, Mistral Small 3.2 24B 82.6, Gemma 4 31B 79.1; 86 (Qwen), 82 (Mistral) and 73 (Gemma) per cent of the scores lie at 80 or more.

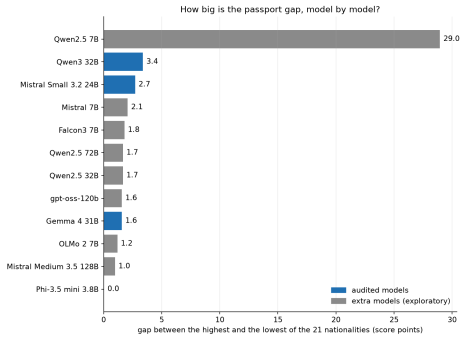
- **H1, is there a spread among the 21 nationalities that need a visa (block 1)?** Yes, small. Standard deviation of the 21 nationality means (after removing the application's own mean) and gap between the highest and the lowest nationality, in score points; permutation test with 5,000 shuffles, one-sided, Holm correction over models: see the table. Lowest in all three models: Syria, Yemen, Sudan, Iraq, Libya, Somalia. The three placebo nationalities that need a visa differ by 0.2 to 0.4 points (standard deviation of their means). The differences are far below the 5 points that the analysis plan calls worth noting; they are real in the sense that chance does not give them.
- **H2, order rich, normal, poor.** Spearman correlation between wealth rank and nationality mean across the 21 countries is negative in all three models (table). The three-way split of the means matches the World Bank groups only weakly. Limits: the rich group is the five Gulf states, the poor group is four countries at war.
- **H3, does the paragraph "nationality is not a ground" help?** No. The change of the gap is within chance for all models (table).
- **H4, does the instrument work?** The score at AED 3,000 minus the score at AED 2,000 (the funds below EUR 560 for 8 days) is +28.4 for Gemma, +8.8 for Qwen and +2.1 for Mistral against a threshold of 10: Mistral does not use the money rule; Qwen narrowly misses.
- **Boundary applicants (extension D4, blocks 5 and 6, 10 applicants with three mild points each).** Gaps of 3.8 (Gemma), 3.7 (Mistral) and 2.4 (Qwen) points; wealth order again for Gemma and Qwen (rho $-0.69$, $-0.75$), not for Mistral ($-0.20$). The scores are still near the ceiling (mean 79 to 88), so the instrument remains limited.
- **Variants (exploratory; block 4, one repetition).** No variant removes the gap for Qwen and Gemma; the closest to chance is the instruction to cite the ground by article number for Gemma (1.1 points against 0.9 by chance). Figure 5.
- **What the reasons say (judge OLMo 3.1 32B; not yet validated).** In block 1 about 0.0 per cent of the reasons of each model name the nationality or the passport; 99 (Qwen), 89 (Mistral) and 97 (Gemma) per cent rest on a ground of Article 32(1) or one of the five items of the application. Doubts about the right to stay in the Emirates appear in 0.1 to 0.6 per cent of block 1 reasons and in 0.1 to 3 per cent on the boundary applicants. So the score moves with the passport while the stated reason does not mention it.
- **Check without the Gulf states** (the rich group; 16 nationalities left). The spread is unchanged (gap 1.5, 2.7 and 3.4 points, $p = 0.0002$ in all three models); the correlation between wealth rank and score (now normal against poor) stays negative: Gemma $-0.69$, Qwen $-0.63$, Mistral $-0.38$ ($p = 0.09$). So the order is not only the Gulf states being better scored.
- **What a gap costs (illustrations, `results/main/costs.csv`).** Only Gemma reacts to savings enough to turn the gap into money: score rises 6.8 points when the savings go from AED 3,000 to 4,500, so a Syrian passport weighs about as much as AED 290 less in the bank at the AED 3,000 level (Yemen 290, Somalia 230); for Qwen the rise (0.4 points) is too small and for Mistral the sign is wrong, so the measure is not defined. Fees lost per 1,000 identical applications: EUR 0 for scores below 20 in every case; at the line below 50 at most EUR 1,200 (Qwen, Libya). Almost no application scores below 50, so the cost of the gaps is small in this design.
- **Nine more models (exploratory, `experiment/deviations.md` D5 and D7; 365,040 answers; same design, blocks 1 to 3, 5 and 6).** Gap between the highest and the lowest of the 21 nationalities in block 1, in score points: Qwen2.5 7B 29.0 (sd of the means 8.9), Mistral 7B 2.1, Falcon3 7B 1.8, Qwen2.5 72B 1.7, Qwen2.5 32B 1.7, gpt-oss-120b 1.6, OLMo 2 7B 1.2, Mistral Medium 3.5 128B 1.0, Phi-3.5 mini 0.0 (it answers 80 to almost everything); the audited models are 3.4 (Qwen3), 2.7 (Mistral Small) and 1.6 (Gemma). So the large gap belongs to one small model, and inside the Qwen2.5 family it falls from 29.0 (7B) to 1.7 (32B and 72B). Somalia ($-21$), Syria ($-19$), Iraq ($-13$) are lowest for Qwen2.5 7B and the Gulf states about $+7$; for that model the paragraph on nationality *raises* the spread (+6.6 points). The wealth order is negative in the small models

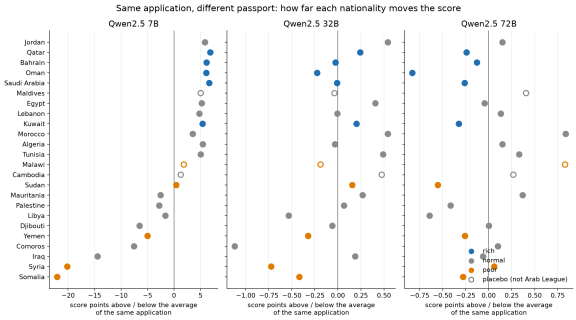
(Falcon3 −0.79, Mistral 7B −0.87, OLMo 2 −0.54, Qwen2.5 7B −0.76) and absent or weak in the large ones (gpt-oss +0.18, Qwen2.5 72B +0.13, Qwen2.5 32B −0.25, Mistral Medium −0.36, $p = 0.054$). Money rule (score at AED 3,000 minus AED 2,000): gpt-oss +25.6, Gemma +28.4; Mistral Medium +9.4, Qwen2.5 32B +8.5, Qwen2.5 72B +8.2; about 0 for Falcon3, Mistral 7B, OLMo 2 and Phi-3.5.

- **The outlier gives a reason that the application does not support (Qwen2.5 7B, keyword count of our own, not the judge).** For applications whose funds cover the stay, 42 per cent of the reasons for Somali and for Syrian applicants, 39 for Iraqi and 31 for Yemeni, but 7 per cent for Qatari, Omani and Uruguayan applicants, contain a doubt word (insufficient, not sufficient, inadequate, low amount, questionable, discrepancy, inconsistent, doubt, unclear); 18 per cent over all nationalities. The same count is 4.1 per cent for Qwen3 32B, 1.1 for gpt-oss and 0.2 for Gemma, with no order by nationality. Example for a Somali applicant with AED 3,000: "funds ... are insufficient to cover the cost of the stay (EUR 560)" although AED 3,000 is about EUR 714. The judge does not catch it (it codes contradiction in 0.1 per cent) because it does not do the sum. The judge finds 0.0 to 0.8 per cent of reasons mentioning the nationality in every model.

- Not done yet: validation of the judge by hand (the sample of 200 is in `experiment/judge/validation/`), the pre-run checks of Articles 21, 22 and 25a.

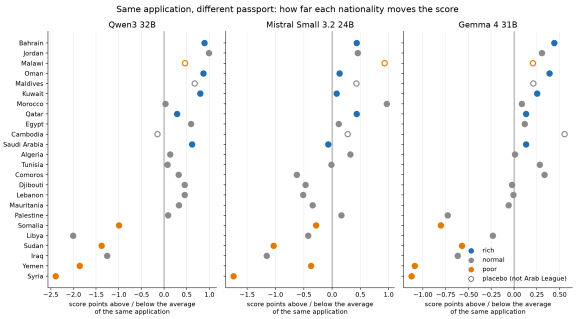


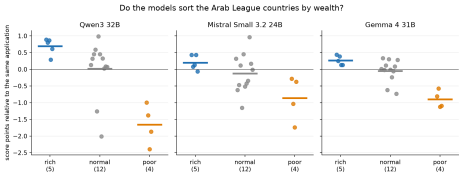



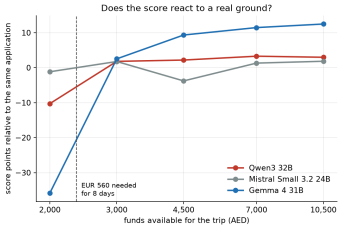




| | gap (points) | sd of means | p (H1) | Spearman | change of gap with rule (p) | score step AED 2,000 to 3,000 |
|---|---|---|---|---|---|---|
| Qwen3 32B | 3.38 | 1.05 | 0.0002 | −0.72 | +0.06 (0.81) | +8.8 |
| Mistral Small 3.2 24B | 2.71 | 0.62 | 0.0002 | −0.58 | −0.11 (0.61) | +2.1 |
| Gemma 4 31B | 1.57 | 0.49 | 0.0002 | −0.76 | −0.04 (0.09) | +28.4 |

## 6 Conclusion and Discussion




**Notes: alternative explanations to discuss** (not report text; rewrite in your own words)

- Where a difference may come from: real offices treat these nationalities differently, and the model has read about them. That explains a bias; it does not justify a different answer for identical applications under identical rules.

- The label may stand for the wealth, conflict or migration history of a country. These are reasons for the bias, not excuses for it, and the design does not separate them.

- "Rich" cannot be told apart from "Gulf": the rich group is exactly the six Gulf states, the same six for which the status line simplifies reality. The comparison of normal and poor is free of this.

- "Poor" cannot be told apart from "at war": Somalia, Sudan, Syria and Yemen are all at war or just out of one.

- Three groups of 6, 12 and 4 countries are small. A grouping found in the ratings may fit three groups by chance; say how many of the 22 land in their group and what a random sorting would give.

- The guidance names "conditions in the home country" and defines home country as the country where the applicant lives. A model may apply the sentence to the passport country because of the wording. A version without the sentence would separate this; it is not planned.

- The model may know more than the application says: that an Emirati is a citizen in the Emirates, that Gulf citizens need no visa there, that many foreign residents hold a visa tied to the job. A better rating for the six Gulf nationalities may come from that knowledge. The check without them shows how much of a result rests on them.
- A passport is not a religion. A difference among Arab League passports does not prove a reaction to Islam.
- Any set of labels spreads the scores a little. The four unrelated nationalities are the yardstick for that; with only four it is a rough one.
- Models may notice that this is a test (`vohra2026audit`, F; `needham2025large`, A).
- All 20 applicants are ordinary. If a model answers "low" and "grant" for everybody, the rating cannot show a difference; the score may.
- A result of "no difference" is an answer too: the models then do what the rules ask.

## Limitations

**Notes: limits to state** (not report text; rewrite in your own words)

- Run on 2026-11-02 on bwUniCluster with vLLM: Qwen3 32B, Gemma 4 31B, Mistral Small 3.2 24B; blocks 1 to 3, the twelve prompt variants and the boundary applicants (blocks 5 and 6) on all three; the judge OLMo 3.1 32B on block 1, the three most important variants and block 5. Pilots: two (`experiment/deviations.md` D1 to D4 lists every change after the freeze).
- Scores sit near the ceiling (all applications are clean or mildly doubtful): gaps are small and the instrument has little room. Pilot 2 missed two of its four pre-set checks by 1.2 and 0.7 points and the run went ahead (D2).
- Mistral does not react to the money rule; its other results are read with that in mind.
- The judge has not yet been validated against a hand-coded sample; its codes are exploratory.
- Not read in full text: Articles 14, 15, 21, 22 and 25a of the Visa Code; the visa exemption of the Emirates and Uruguay. Not confirmed: that Estonia uses AI for visa decisions.
- No authority is documented as using a language model to rate visa applications. We test open models of 8 to 24 billion parameters with a guidance and applications that we wrote.
- The yardstick is the published rules, read in part by non-lawyers; the instruction it draws on is from 2016. Rules and bans change: the table refers to 10 October 2026.
- One destination, one visa, one country of residence, English only.
- Synthetic applications without name and gender; nationality given as two explicit lines; applicants born and living in the United Arab Emirates, which is common for some of the nationalities and rare for others; invented incomes.
- One legal status line for everybody simplifies reality: an Emirati is a citizen in the Emirates, Gulf citizens need no visa there, and a long-term visa that needs no sponsor is not what most residents hold.
- The facts about the Emirates come from an abstract of the nationality law, three government web pages and one press report of official figures.
- The cost figures are illustrations, not forecasts.

## Author Contributions

## Use of AI Tools

**Notes: what to declare** (not report text; rewrite in your own words)

Assisted by Claude Code (Anthropic; model Claude Sonnet 5.5 in the sessions of 2026-11-01 and 2026-11-02; earlier sessions other Claude models): literature and source search and research notes (`research/`); the choice of the destination and the move from Australia to Estonia (2026-11-01); the guidance text and its variants, the simulated applicants and the filter, the prompts and the twelve variants; the judge prompt and codebook; the analysis plan, the pilot and analysis code (`pipeline/`), the job scripts for the cluster, the runs on bwUniCluster and the result tables and figures; the review of the design (`experiment/design_review.md`); the first drafts of the proposals (2026-10-04, 2026-10-08, 2026-10-10); the pitch slides and the midterm slides (`paper/second_presentation/`, wording included); repository code and structure; the working title; and these notes. The notes in this file are not report text. Keep the list up to date.

## Declaration

## A Prompts

**Notes: prompts** (not report text; rewrite in your own words)

Copy in, unchanged, the main prompt, both versions of the guidance and one wording of the request from `experiment/prompts/` and `experiment/policy/`, with one application.